Notes: Goldin & Katz (JPE, 2002)

The Power of the Pill: Oral Contraceptives and Women's Career and Marriage Decisions

Contents

1 Big picture

- **Question.** Did the diffusion of the oral contraceptive pill among young unmarried women change women's professional investments, marriage timing, and related life-cycle decisions?
- **Main answer.** Yes. The pill lowered the direct cost of delaying marriage and investing in long-duration professional education, and also changed the delayed-marriage market.
- **Key puzzle.** The pill was approved in 1960 and spread rapidly among married women, but young single women did not use it widely until the late 1960s and early 1970s because legal and social barriers restricted access.
- **Source of empirical variation.** States changed age-of-majority laws and mature-minor rules at different times. Cohorts reached ages 17–20 under different legal regimes.
- **Economic takeaway.** A reproductive technology can change human-capital investment and marriage-market equilibria when it changes the cost of sex, marriage, and fertility timing.

2 Data and measurement

2.1 Legal and historical data

Tables 1 and 2 organize the legal chronology and state-by-state access rules. They distinguish invention from access. FDA approval in 1960 did not immediately make the pill available to young unmarried women.

Interpretation: The legal data are the treatment map for the empirical design.

2.2 Pill-use data

The paper uses the National Survey of Young Women 1971, the National Survey of Adolescent Female Sexual Behavior 1976, the National Health Interview Survey 1987, and the National Survey of Family Growth 1982.

Interpretation: No single survey contains all outcomes, so the paper builds a collage of evidence.

2.3 Career, marriage, and sexuality outcomes

Career outcomes include professional-school enrollment and adult professional occupations. Marriage outcomes include age at first marriage and marriage before age 23. Sexual behavior outcomes include premarital sex and first family-planning visit timing.

Interpretation: These outcomes correspond directly to the theory: the pill lowers the cost of delayed marriage and long-duration career investment.

3 Models and empirical specifications

3.1 Two-period framework

$$\text{If marry in period 1: woman receives } Y_i, \quad \text{man receives } N_j.$$
$$\text{If delay and invest: woman receives } Y_i + a_j - \lambda, \quad \text{man receives } N_j + a_j - \lambda.$$

Interpretation: The parameter a_j is career ability and λ is the cost of delaying marriage. The pill lowers λ.

3.2 Career cutoff and marriage-market value

$$\text{Career investment and delayed marriage occur when } a_j \geq \lambda.$$
$$F_j = \max\{N_j, N_j + a_j - \lambda\}.$$

Interpretation: Lower λ shifts the career cutoff left and raises the marriage-market value of high-career-ability women.

3.3 Empirical law-access models

$$PillUse_{is} = \alpha + \beta NonrestrictiveLaw_s + X_i'\gamma + Region_s + \epsilon_{is}.$$
$$1\{Married < 23\}_{isc} = \alpha + \beta PillAccess_{sc} + \delta AbortionAccess_{sc} + State_s + Cohort_c + X_i'\gamma + \epsilon_{isc}.$$

Interpretation: Table 3 is the law-to-pill-use first stage; Tables 4 and 5 study marriage and careers.

4 Identification and empirical design

The key identification comes from state-by-cohort variation in legal access to contraception for young unmarried women. Some states lowered the age of majority or adopted mature-minor doctrines earlier than others, and women born in different years reached ages 17–20 under different legal regimes. The main threats are state-specific liberalism, abortion reform, feminism, antidiscrimination law, migration, and measurement error. The paper addresses these with controls, abortion variables, state/cohort variation, and robustness checks.

5 Section-by-section study guide

- **The Pill and Single Women.** Explains delayed diffusion among unmarried women and the legal basis of access variation.
- **Framework.** Models the direct and indirect effects of the pill on career and marriage.
- **Time series.** Documents turning points in pill use, professional school entry, marriage, and premarital sex.
- **Econometrics.** Uses state-cohort law variation to estimate marriage and career effects.
- **Alternatives.** Evaluates abortion, feminism, antidiscrimination, draft deferments, and sex ratios.

6 Tables and figures with expanded interpretation

6.1 Table 1: Landmark events

Read: FDA approval came in 1960, but key access changes for unmarried minors came later. Griswold protected married couples; later mature-minor and age-of-majority changes affected single young women.

Detailed interpretation: Table 1 is both a historical chronology and an identification tool. It explains why FDA approval is not the relevant treatment date for college women.

Economic message: Technology affects behavior only when institutions make it usable.

TABLE 1
LANDMARK EVENTS AND DECISIONS REGARDING THE PILL AND CONTRACEPTIVE USE

Year	Event or Landmark Decision
1873	Congressional passage of the Act for the Suppression of Trade in, and Circulation of Obscene Literature and Articles of Immoral Use, also known as the "Comstock Law"
1916	In direct violation of state law, Margaret Sanger opens the first birth control clinic in the United States
1936	U.S. Circuit Court of Appeals modifies the Comstock Law to allow the dissemination of birth control information and devices
1937	Researchers discover the function of progesterone in inhibiting ovulation
1949	Russell Marker synthesizes inexpensive cortisone for the treatment of rheumatoid arthritis
1950	Margaret Sanger convinces the heiress Katherine Dexter McCormick to fund research on "the pill"
1951	Carl Djerassi at Syntex synthesizes norethindrone, an orally active progestational hormone
1952	Chemist Frank Colton at G. D. Searle develops norethynodrel, chemically similar to norethindrone
1953	Katherine Dexter McCormick promises Gregory Pincus (researcher and Searle consultant) to fund his project to develop a birth control pill through its completion
1954	Researchers John Rock and Gregory Pincus conduct the first tests using norethynodrel and norethindrone to prevent ovulation
1955	Searle's Frank Colton awarded a patent on norethynodrel
1956	Large-scale trials begin to assess the drug's effectiveness as a contraceptive
1956	Syntex's Carl Djerassi awarded a patent on norethindrone
1957	The FDA approves Syntex's Norlutin (norethindrone) and Searle's Enovid (norethynodrel) for treatment of hormonal and other medical disorders
1960	The FDA approves the use of norethynodrel (Enovid) as an oral contraceptive for women
1961	The first reports of thromboembolism attributed to the pill surface in Britain
1962	Ortho Pharmaceutical Corp. enters the oral contraceptive market
1963	The FDA concludes that there is no connection between Enovid and thromboembolism
1964	Parke-Davis and Syntex enter the oral contraceptive market
1965	The U.S. Supreme Court in *Griswold v. Connecticut* overturns a Connecticut law prohibiting the use of contraceptives on the grounds that it violates a married couple's right to privacy
1968	Papal encyclical is issued enjoining Catholics from using the pill
1969	Yale University opens one of the first college family planning clinics with access to all students
1970	Senator Gaylord Nelson holds hearings investigating the health risks of oral contraceptives
1970	The FDA requires informational pamphlet on health risks in every package of birth control pills
1971	July 1, 1971, the 26th Amendment is ratified; most states also lower the "age of majority"
1972	The U.S. Supreme Court in *Eisenstadt v. Baird* overturns the Massachusetts law prohibiting the sale of contraceptives to unmarried persons
1974	In Wisconsin Federal District Court, *Baird v. Lynch* overturns a law prohibiting the sale of contraceptives to unmarried individuals (Civ. No. 71-C-254, W.D. Wis.)

Table 1: Landmark events

6.2 Figures 1 and 2: Pill use and family-planning services

Read: Pill use before young adult ages rises sharply for cohorts born around 1948–1952. First family-planning visits before age 21 follow the same pattern.

Detailed interpretation: The figures show that pill use shifted earlier in the life cycle, precisely when women were making college, marriage, and career decisions.

Economic message: Early access turns the pill into a human-capital investment technology.

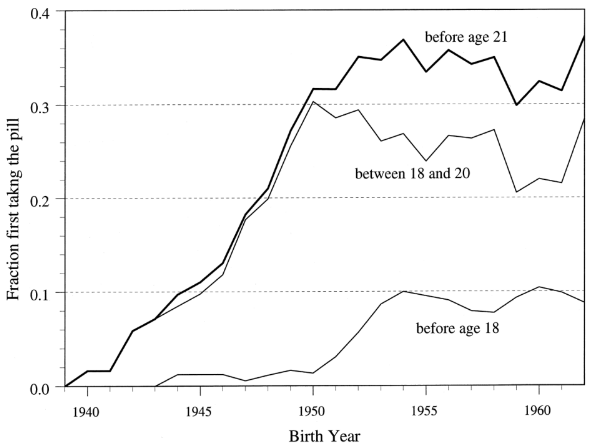


FIG. 1.—Fraction of college graduate women first taking the pill at various ages (among those with no births before age 23). Source: Inter-university Consortium for Political and Social Research (1990). Three-year centered moving averages are shown.

(a) First pill use by age and cohort

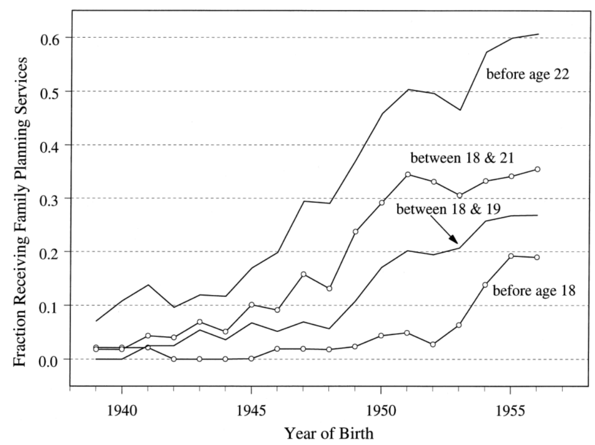


FIG. 2.—Fraction of college graduate women receiving first family planning services at various ages (among those not married by age 22). Source: Inter-university Consortium for Political and Social Research (1985). Three-year centered moving averages are shown.

(b) First family-planning services

6.3 Table 2: State laws

Read: Table 2 maps state laws on contraceptive access and age of majority from 1969 to 1974.

Detailed interpretation: This table is the treatment-assignment matrix for the state-cohort design.

Economic message: Legal heterogeneity converts national social change into empirical variation.

TABLE 2
STATE LAWS REGARDING CONTRACEPTIVE SERVICES TO MINORS AND THE AGE OF MAJORITY, 1969–74

	Age of Majority			Earliest Legal Age to Obtain Contraceptive Services without Parental Consent		
State	1969 (1)	1971 (2)	1974 (3)	1969 (4)	1971 (5)	1974 (6)
Ala.	21	21	21	21	17	17
Alaska	19	19	19	19	19	14 or 19*
Ariz.	21	18	18	21	18	18
Ark.	18†	18†	18†	18	14	14
Calif.	21	21	18	15	15	15
Colo.	21	21	21	21	14	14
Conn.	21	21	18	21	18	18
Del.	21	21	18	21	21	18
D.C.	21	21	21	21	14	14
Fla.	21	21	18	21	21	14
Ga.	21	21	18	14	14	14
Hawaii	20	20	20	20	20	20
Idaho	18†	18†	18	18	18	14
Ill.	21	18	18	21	14	14
Ind.	21	21	18	21	21	18
Iowa	21	21	18	21	21	14 or 18*
Kans.	21	21	18	21	21	14
Ky.	18	18	18	18	18	14 or 18*
La.	21	21	18	21	21	14
Maine	21	18	18	21	18	18
Md.	21	21	18	21	18	14
Mass.	21	21	18	21	21	18
Mich.	21	21	18	21	14	14
Minn.	21	21	18	21	18	18
Miss.	21	21	21	14	14	14
Mo.	21	21	21	21	21	21
Mont.	21	19	18	21	19	18
Nebr.	20	20	19	20	20	19
Nev.	18†	18†	18	18	18	18
N.H.	21	21	18	21	14	14
N.J.	21	21	18	21	21	18
N.M.	21	18	18	21	18	14 or 18*
N.Y.	21	21	18	21	16	16
N.C.	21	18	18	21	18	18
N.D.	21	18	18	21	18	18
Ohio	21	21	18	21	21	14
Okla.	18†	18†	18	18	18	14 or 18*
Ore.	21	21	18	21	15	15
Pa.	21	21	21	21	18	18
R.I.	21	21	18	21	21	18
S.C.	21	21	21	21	21	16
S.D.	21	21	18	21	21	18
Tenn.	21	18	18	21	14	14
Texas	21	21	18	21	21	18
Utah	18†	18†	18†	18	18	18
Vt.	21	18	18	21	18	18
Va.	21	21	18	21	21	14
Wash.	21	18	18	21	18	18
W.Va.	21	21	18	21	21	14 or 18*

(a) State laws, part 1

TABLE 2
(*Continued*)

	Age of Majority			Earliest Legal Age to Obtain Contraceptive Services without Parental Consent		
State	1969 (1)	1971 (2)	1974 (3)	1969 (4)	1971 (5)	1974 (6)
Wisc.	21	18	18	21	18	18
Wyo.	21	21	19	21	21	14 or 19*
Summary	States <20: 7	States <20: 18	States <20: 43	States ≤16: 3	States ≤16: 12	States ≤16: 27‡

SOURCE.—1969: Pilpel and Wechsler (1969). 1971: U.S. Department of Health, Education, and Welfare (1974). 1974: Paul et al. (1974). The coding of the laws is as of June 1974. 1974: U.S. Department of Health, Education, and Welfare (1978).

NOTE.—"Obtaining contraceptive services" means the ability to get the birth control pill. Some states had a different age for voluntary sterilization, and abortion laws occasionally differed as well. We use an age of 14 when the law was interpreted to mean that any minor could receive contraceptive devices without parental consent. Ala.: Law allows high school graduates and married women to obtain contraception, but not any female under the age of majority; therefore, an age of 17 is coded. Ark.: Age of majority is 18, but recent law allows all women to get state family planning services except single women at college away from home, who should go to a private doctor. Conn.: Age of majority was reduced to 18 in 1972. An earlier law enabled any minor of 18 or older to obtain health services. Kans.: Legislation in 1966 allowed a physician to prescribe birth control to any woman at a public clinic, but the law was not universal. See, however, the discussion in Bailey (1997), which discusses the case of Lawrence, Kans. Ky.: A "mature minor" decision was effective in 1972. Maine: Physician must find "probable health hazard"; thus the law is not universal. Md.: Age of majority is 21, but consent rights are given to those 18 years and older for contraceptive services. Mich.: Age of majority is lowered to 18 in 1972. Minn.: Law states that minors living apart from their parents can give consent to health services; thus the age in 1971 is given as 18 since it does not provide blanket coverage for 14–17-year-olds. N.J.: Age of majority is lowered to 18 in 1973. N.Y.: Pharmacists may sell contraceptives to minors 16 and older. The law was ambiguous with regard to physicians and parental consent. Ohio: Ohio has a mature minor doctrine established in 1956, but its relevance to birth control is unclear in 1971. Pa.: Minors 18 years and older (or high school graduates) may consent to any medical care. Va.: Age of majority is lowered to 18, and any individual under age 18 may consent to birth control services, except abortion and sterilization, effective July 1972. Wisc.: The restriction on contraceptives to unmarried individuals was put in question after the *Eisenstadt v. Baird* (1972) decision regarding a similar Massachusetts law.

* The state has a comprehensive family planning program that does not exclude the provision of contraceptive services to minors, but there is either no mature minor doctrine in the state or no clear decision by the state attorney general concerning the legality of such a provision.

† Age of majority is 18 for females and 21 for males.

‡ Seven states are ambiguous cases.

extensions of the mature minor doctrine. Of the 21 states that changed their regulations between 1969 and 1971, 10 reduced the age of majority, 10 instituted a mature minor doctrine, and one instituted a family planning act.[22]

Some ambiguity surrounds the meaning of these legal changes. In no state was it per se illegal in 1972, for example, to prescribe or sell contraceptives to a minor.[23] Rather, the legality both before and after

[22] The 1967 amendment to Title IV of the Social Security Act (Aid to Families with Dependent Children) requires state and local welfare agencies to provide contraceptive services to eligible individuals without regard to marital status and age. But this law would not have affected the ineligible population, and it is unclear how it affected teenagers in states without a mature minor doctrine.

[23] Until 1966 it was illegal in Massachusetts to sell contraceptives. The law was modified after *Griswold v. Connecticut* (1965). From 1966 to 1972, it was illegal to sell contraceptives to unmarried persons, and sale to married persons required a prescription. That law was struck down by the U.S. Supreme Court in *Eisenstadt v. Baird* (March 1972). A similar Wisconsin law was overturned in *Baird v. Lynch* (1974, Civ. No. 71-C-254, W.D. Wis.). See U.S. Department of Health, Education, and Welfare (1974, 1978). There is no evidence that these laws were rigorously enforced.

(b) State laws, part 2

6.4 Table 3: State laws and pill use

Read: Nonrestrictive laws raise pill use, especially for older teenagers and college-attending women.

Detailed interpretation: This is the behavioral first stage linking legal access to actual use.

Economic message: The legal instrument matters for the target population.

TABLE 3
STATE LAWS AND PILL USE AMONG NEVER-MARRIED FEMALE YOUTHS IN THE NATIONAL SURVEY OF YOUNG WOMEN, 1971
Dependent Variable: 1 = Ever Taken the Birth Control Pill

	15–19 Years Old		17–19 Years Old		17–19 Years Old and Attends College	
	All (1)	Sexually Active* (2)	All (3)	Sexually Active (4)	All (5)	Sexually Active (6)
Mean of dependent variable	.0652	.245	.106	.304	.145	.406
State law (1=nonrestrictive for minors)	.0232	.0807	.0422	.109	.0701	.128
	(.00870)	(.0280)	(.0153)	(.0372)	(.0367)	(.0889)
Age variables:						
15	−.140	−.259				
	(.210)	(.0641)				
16	−.119	−.175				
	(.0186)	(.0521)				
17	−.0946	−.106	−.0730	−.0837	−.114	−.103
	(.0163)	(.0443)	(.0202)	(.0470)	(.0491)	(.146)
18	−.0618	−.0941	−.0544	−.0836	−.0602	−.0632
	(.0618)	(.0334)	(.0170)	(.0373)	(.0293)	(.0664)
Education variables:						
College			.0680	.177		
			(.0177)	(.0430)		
Currently attends school	−.0820	−.0695	−.0714	−.0845	−.211	−.323
	(.0158)	(.0364)	(.0189)	(.0405)	(.0846)	(.161)
Catholic	−.0126	−.0272	−.0210	−.0426	−.0256	−.0804
	(.00845)	(.0287)	(.0149)	(.0374)	(.0334)	(.0819)
African-American	.0775	.0656	.101	.0405	−.0276	−.186
	(.0112)	(.0284)	(.0193)	(.0379)	(.0505)	(.0958)
Observations	4,211	1,314	2,226	890	647	245
R^2	.0854	.113	.0645	.0856	.0558	.103

SOURCE.—National Survey of Young Women, 1971.
NOTE.—Standard errors are in parentheses. Ordinary least squares estimates are given and are almost identical to the implied slopes estimated by a logit and evaluated at the sample means. All regressions use the NSYW71 sampling weights. State law is coded one if, in 1971, the state allowed minors (16 years and older) to receive birth control services and devices without parental consent. There were 12 states in that group; see col. 5 of table 2. Dummy variables are included in cols. 1 and 2 for grades 8 and above, and dummy variables are included for census divisions in all columns. About 3 percent of the sample in cols. 5 and 6 were not attending college at the time of the survey but reported that they attended college.
* "Sexually active" means that the individual had ever had sexual intercourse.

(see Bailey [1997] for a case study of Kansas). Thus pill use by young, single women began to rise about a year before the sea change in the laws regarding the age of majority.

II. Frameworks to Understand the Effect of the Pill on Marriage and Career

Table 3: State laws and pill use

6.5 Figure 3: Theoretical model

Read: Lowering λ from λ_0 to λ_p moves the career cutoff left and creates Group II, women induced to delay marriage and invest.

Detailed interpretation: The model distinguishes direct effects, marriage-market ranking effects, and possible losers among low-career-ability women.

Economic message: The pill changes relative prices in career and marriage decisions.

6.6 Figure 4: Female professional students

Read: Womenś professional-school entry rises sharply around 1970 across law, medicine, dentistry, and business.

Detailed interpretation: The timing lines up with pill access among young unmarried women and the professions require long up-front investment.

Economic message: The pill lowered the cost of long-duration professional investment.

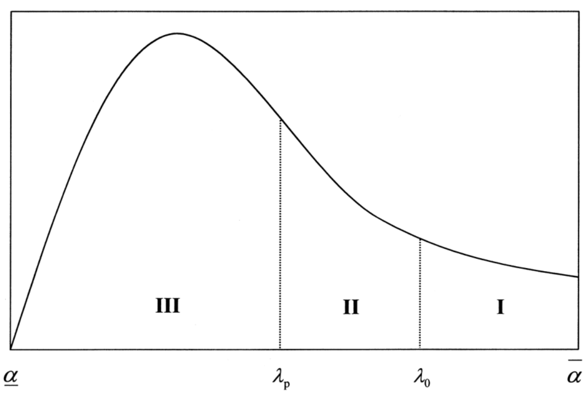
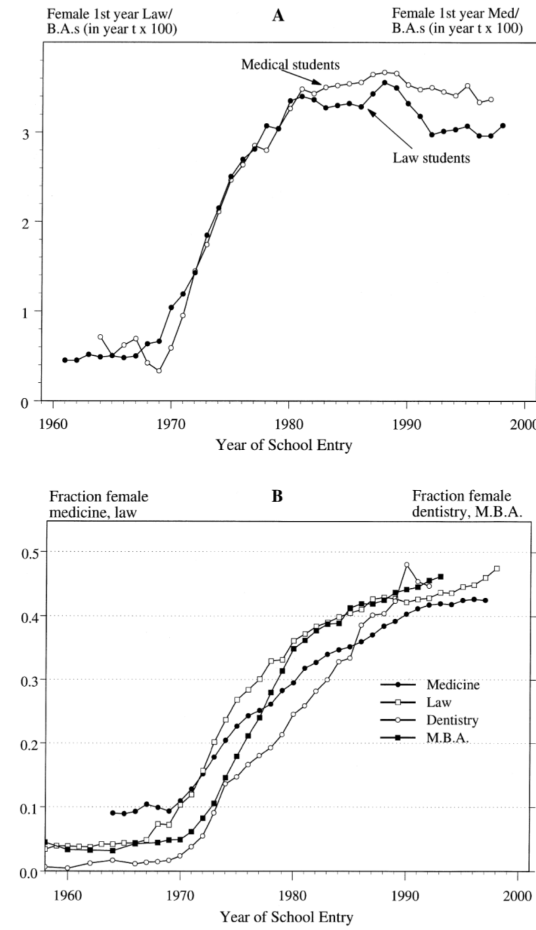


FIG. 3.—Distribution of career ability, α, and shifts in the "impatience" factor, λ, with the pill.



. 4.—First-year female professional students as a percentage of female B.A.'s
d as a fraction of first-year students (panel *B*). Source: B.A. degrees: U.S. Depa
ucation (1998), table 244. First-year medical students: *Journal of the American*
ation (various years 1978–98). First-year law students: American Bar Associati
http://www.abanet.org/legaled/femstats.html). First professional degrees
U.S. Department of Education (1998), table 259. Earned degrees in busine
rtment of Education (1997), table 281. Note: Data for first-year dental and b
nts are derived from first professional degrees lagged four years for dental s
hree years for business students. The data, for years of overlap, are similar t
st-year students from *Students Enrolled for Advanced Degrees* (U.S. Department of
tion, and Welfare, various years). The procedure, moreover, produces values
se for medicine and law for which the first-year student time series exists.

6.7 Figures 5 and 6: Marriage and premarital sex

Read: Early marriage declines for college women born around 1950 and after, while premarital sex rises.

Detailed interpretation: These figures show that sex, marriage, and career timing were reorganized.

Economic message: Delayed marriage became feasible without requiring delayed sex.

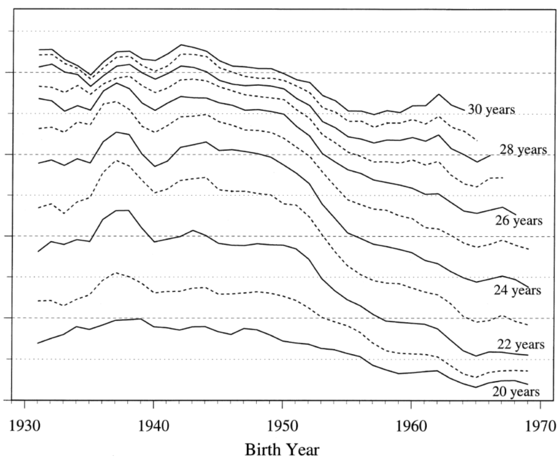


Fig. 5.—Fraction of college graduate women married before various ages. Source: Current Population Survey, Fertility and Marital History Supplement, 1990 and 1995. Three-year centered moving averages are shown.

(a) Marriage before ages

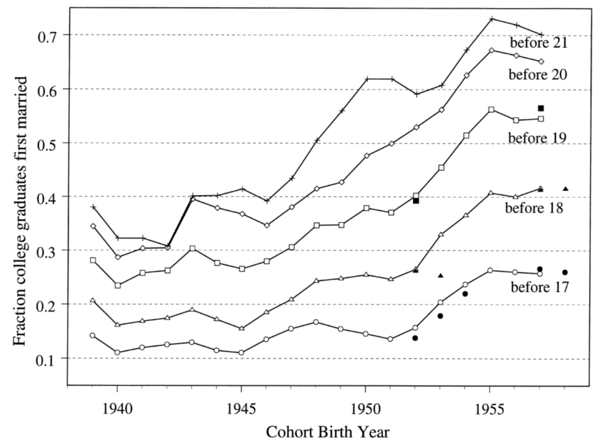


Fig. 6.—Fraction of never-married women having sex before various ages. Source: All but the solid markers: Inter-university Consortium for Political and Social Research (1985). Solid markers for birth cohorts of 1952, 1953, and 1954: Zelnik and Kantner (1989). Solid markers for birth cohorts of 1957 and 1958: Inter-university Consortium for Political and Social Research (1982). Three-year centered moving averages are shown. Solid markers, of the same shape as the open markers, give the values for contemporaneous data.

(b) Premarital sex before ages

6.8 Table 4: State laws and age at first marriage

Read: Earlier pill access reduces the probability of marriage before age 23.

Detailed interpretation: The table moves from time-series coincidence to state-cohort causal evidence.

Economic message: Earlier pill access delayed marriage among college women.

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TABLE 4
State Laws and the Age at First Marriage for College Women (U.S. Natives Born 1935–57)
Dependent Variable: 1 = Married before Age 23

	College Graduates							Some College or More	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean of dependent variable	.41	.41	.41	.41	.41	.41	.41	.53	.53
Nonrestrictive birth control law at age 18*	−.0196	−.0162	−.0207	−.00986	−.0227			−.0124	
	(.00737)	(.00762)	(.00920)	(.00791)	(.00917)			(.00600)	
	[.0109]	[.0105]	[.00941]	[.0107]	[.00995]			[.0100]	
Pill access by age 17†						−.0262	−.0324		−.0240
						(.0115)	(.0131)		(.00872)
						[.0163]	[.0143]		[.00143]
Pill access by ages 18–20‡						−.00894	−.0126		−.0132
						(.00822)	(.00821)		(.00593)
						[.00922]	[.00920]		[.00676]
Legalized abortion at age 18§		−.0236	−.0114					−.00974	−.00904
		(.00992)	(.00956)					(.00777)	(.00761)
		[.0103]	[.0103]					[.00727]	[.00705]

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TABLE 5
Impact of Pill Access and Abortion Legalization on Career and Marital Status for College Women, 30–49 Years Old
(U.S. Natives Born 1921–60)

	Professional Occupation, Excluding Teachers, Nurses		Lawyer, doctor		Never Married	Currently Married	Currently Divorced	Currently Divorced/Ever Married	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean of dependent variable	.127	.127	.0141	.0141	.126	.749	.0916	.104	.104
Fraction using pill before age 21*	.0480		.0352					−.0325	
	(.0275)		(.00539)					(.0257)	
Fraction with pill access law before age 21†		.00410		.0159	.0608	−.00813	−.0596		−.0558
		(.0142)		(.00353)	(.0189)	(.0230)	(.0107)		(.0121)
Average abortion rate from ages 18 to 21‡	.0457		.0306					−.149	
	(.0230)		(.00451)					(.0215)	
Fraction with legalized abortion at age 18§		.0236		.00255	.0431	−.0299	−.0127		−.0135
		(.0146)		(.00362)	(.0194)	(.0236)	(.0110)		(.0124)

6.9 Table 5: Career and marital status

Read: Early pill access predicts professional occupations and changes in marital outcomes at ages 30–49.

Detailed interpretation: This table links access in young adulthood to long-run adult outcomes.

Economic message: Reproductive control enabled long-horizon human-capital investment.

6.10 Table 6 and Table A1

Read: Pill use far exceeds abortion use among 18–19-year-old college women. Table A1 lists the data sets used.

Detailed interpretation: Table 6 addresses alternative explanations; Table A1 explains why the paper must combine many surveys.

Economic message: The pill was the pervasive preventive technology; abortion was complementary but less widely used.

TABLE 6
ABORTION AND PILL USE AMONG 18- AND 19-YEAR-OLD WOMEN, 1971 AND 1976

			Never Married	
	All	Nonvirgins	All	Nonvirgins
	A. Woman 18 and 19 Years Old Attending College in 1971			
As of 1971:				
% ever had an abortion	2.2	5.3	2.0	5.3
	(599)	(253)	(568)	(222)
% ever took the pill	17.4	42.2	15.5	41.3
	(595)	(253)	(564)	(222)
	B. Woman 18 and 19 Years Old Attending College in 1976			
As of 1976:				
% ever had an abortion	3.7	7.2	3.0	6.0
	(177)	(100)	(160)	(87)
% ever took the pill	32.5	61.7	29.0	58.3
	(177)	(100)	(160)	(87)

SOURCE.—National Survey of Young Women, 1971; National Survey of Adolescent Female Sexual Behavior.
NOTE.—Numbers of observations are in parentheses.

men two to three years older, women born early in the baby boom (the mid to late 1940s) should have faced poorer marriage market conditions than those born in the 1950s and 1960s. Yet marriage rates for college women did not increase until cohorts born from the late 1940s to early 1950s and did not decrease for later ones facing more favorable sex ratios.[54]

The most difficult supply-side explanation to assess is the resurgence of feminism in America. Feminism empowered young women to see themselves as the equal of their male peers, and it was complementary to the pill by increasing the number of young women who believed they could aim for the top.

Demand-side explanations involving a change in the relative demand for women by employers and educational institutions can also be offered. They include sex discrimination legislation and the ending of Vietnam-era draft deferments.

Although the Civil Rights Act of 1964 covered discrimination by "sex," the Equal Employment Opportunity Commission (EEOC) set up to investigate charges did little about sex discrimination until the early 1970s.[55] Affirmative action in the form of Executive Order 11246 was amended in 1967 by Executive Order 11375 to cover women, but it took

[54] An interaction between marriage market effects, driven by sex ratio changes, and the pill may help account for this anomaly (Heer and Grossbard-Shechtman 1981).
[55] The National Organization for Women was formed in 1966 to pressure the EEOC to consider discrimination by sex. Sex discrimination complaints rose from about 3,150 in 1970 to 18,150 in 1973, peaking at 29,450 in 1976 (Goldin 1990, fig. 7.1).

(a) Table 6: Abortion and pill use

TABLE A1
DATA SETS CONTAINING INFORMATION ON CONTRACEPTIVE USE, 1955–87

Data Set, Year(s), and Observations	Sample Characteristics
National Fertility Surveys (NFS): 1965 (N=5,617), 1970 (N=6,752), and 1975 followup (N=3,403)	Currently married women, married prior to age 25; retrospective information on contraceptive use was taken only for periods in which respondent was married
National Survey of Young Women (NSYW71): 1971 (N=4,611)	Young women, never married and ever married, 15–19 years old; contains current state of residence
National Survey of Adolescent Female Sexual Behavior (NSAF76): 1976 (N=2,193)	Young women, never married and ever married, 15–19 years old; does not contain current state of residence
National Health Interview Survey (NHIS): 1987 (N=12,747)	Women born 1929–69 (aged 18–67); contains information on age at first pill use, but none on age at first marriage
National Survey of Family Growth, Cycle III (NSFG): 1982 (N=7,969)	Women 15–44 years old; contains information on age at first family planning visit and age at first marriage

[57] See Goldin (1997, app. table 2.1) for data based on the Current Population Survey, series P-20. The fraction graduating from college or university is about 28 percent for cohorts born in the late 1940s.

(b) Table A1: Data sets

7 Short summary

- The pill changed womenś career and marriage decisions by lowering the cost of delaying marriage and investing in professional training.
- State law changes generated variation in access for young unmarried women.
- Earlier access raised pill use, delayed marriage, and increased professional career outcomes.

- Abortion and feminism were complementary, but the pill had broader use, earlier timing, and a more direct channel for professional investment.

8 Expanded conclusion

8.1 Core conclusion

The pill was a major force behind the rise of professional careers and delayed marriage among college-educated women.

8.2 Direct mechanism

The pill reduced pregnancy risk during professional investment years.

8.3 Indirect mechanism

The pill thickened the delayed-marriage market and lowered the marriage-market cost of career investment.

8.4 Evidence interpretation

The paperś evidence is cumulative: legal chronology, pill-use first stage, time-series turning points, marriage regressions, career regressions, and alternative-explanation analysis all point in the same direction.

8.5 Limitations

State legal changes are not randomized, and the evidence is strongest for college-educated women and professional careers.

8.6 Takeaway

Reproductive autonomy changed the timing of sex, marriage, fertility, and human-capital investment.